Marmara University
Engineering Faculty
Electrical and Electronics Engineering
EE4077: Fundamentals of Machine Learning
Midterm Exam Solutions (Spring, 2023)

1.

a) 6 for $P(X|Y)$, 1 for $P(Y)$ (5)

b)

	$Y=1$	$Y=0$
Red?	$\frac{3}{4}$	0
Affordable?	$\frac{1}{2}$	$\frac{4}{5}$
Fuel-Efficient?	$\frac{3}{4}$	$\frac{3}{5}$

(5)

c)

$$P(Y=1|\text{red, not affordable, efficient}) \propto \frac{4}{9}\cdot\frac{3}{4}\cdot\frac{2}{4}\cdot\frac{3}{4}=\frac{1}{8}$$

$$P(Y=0|\text{red, not affordable, efficient}) \propto \frac{5}{9}\cdot 0\cdot\frac{1}{5}\cdot\frac{3}{5}=0$$

(5)

(5)

d) If the car is red, the classifier will always predict I'm interested because $P(\text{not red}|Y=0)=0$. We can use a prior which prevents parameter estimates from being 0, i.e. adding 1 fake count for each feature/label combination.

2.

(a)

$$J(\boldsymbol{\theta}) = -\tfrac{1}{3}[\log(\sigma(3)) + \log(\sigma(-1)) + \log(1-\sigma(-5))] \approx 0.46$$

(5)

(b)

$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_1} = (\sigma(3)-1)1 = -0.05$$
$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_2} = (\sigma(3)-1)2 = -0.10$$
$$\frac{\partial J^{(1)}(\boldsymbol{\theta})}{\partial \theta_3} = (\sigma(3)-1)1 = -0.05$$

(5)

(c)

$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_1} = (\sigma(-1)-1)1 = -0.75$$
$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_2} = (\sigma(-1)-1)1 = -0.75$$
$$\frac{\partial J^{(2)}(\boldsymbol{\theta})}{\partial \theta_3} = (\sigma(-1)-1)-1 = 0.75$$

(5)

(d)

$$\begin{bmatrix}-2\\2\\1\end{bmatrix} - 1\begin{bmatrix}-0.75\\-0.75\\0.75\end{bmatrix} = \begin{bmatrix}-1.25\\2.75\\0.25\end{bmatrix}$$

(5)

(e) J=0.09, it should decrease for logistic classifier to learn. (5)

(f)

$$\sigma(\boldsymbol{\theta}^T X) > 0.5 \implies Y = 1$$

(5)

3.

(a) -[(1/2)log(1/2) + (1/2)log(1/2)] = 1 bit (6)

(b)

$$1 - 0.8 \cdot (-5/8 \cdot \log_2(5/8) - 3/8 \cdot \log_2(3/8)) = 0.23645$$ (6)

(c)

$X_1 < 2$ and $X_1 \geq 2$ (6)

(d)

$0.2 + -0.001$ (6)

(e) Yes, it is possible. (6)

4.

(a)

Curve (ii) is the training set. Training error increases as the training set increases in size (more points to account for). However, the increase tapers out when the model generalizes well. Evidently, curve (i) is testing, since larger training sets better form generalized models, which reduces testing error. (5)

(b) overfitting (5)

5. (a) Regression, classification, clustering, coding/dimensionality reduction (5)

(b) Estimating the price of a house based on a dataset is a regression problem. Deciding if an email is spam or not based on a dataset is a classification problem. (5)